

Computer Science & IT

Database Management System



Query Languages

Lecture No. 02



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Recap of Previous Lecture

- ✓ Topic Query Languages
- ✓ Topic Relational algebra
- ✓ Topic Basic relational algebra operations



Topics to be Covered



✓ Topic

Derived relational algebra operations

✓ Topic

Different types of join operations

★ Topic

Division operation





Topic : Query languages

Query languages

Procedural query language

We define, what needs to be retrieved from the database, along with the procedure to retrieve the data from database

eg: Relational Algebra

Non-procedural query language

We only define, what needs to be retrieved from the database, and we use the syntax provided by that query language to retrieve the information from the database

* eg: SQL
Tuple Relational Calculus (TRC)
Domain Relational Calculus (DRC)

Note:-

Query condition evaluates tuple by tuple, only
One tuple at a time.

→ ∴ If we want to compare two or more tuples of the same table or different tables, then we need to Join those tuples into a single tuple

↓
To join the tuples
we can use

- ↙ Cross Join (or)
- ↓ Natural Join
- ↘ Some other type of Join opt.

Note:- Relational algebra query will always produce distinct tuples.



Topic : Relational Algebra

- ✓ Relational Algebra is a procedural query language used to query the relational database tables to access data.
- * Relational Algebra operation can be classified into two types:
 - ✓ 1) Basic Relational Algebra Operations
 - ✓ 2) Derived Relational Algebra operations



Topic : Basic Relational Algebra operators

- ✓ 1. Projection(π) ✓
 - ✓ 2. Selection (σ) ✓
 - ✓ 3. Cross Product ($\times$) ✓
 4. Union (U) ✓
 5. Set Difference ($-$) ✓
 6. Rename (ρ) ✓
- we have already discussed
when we were discussing
"natural join" oph.
- Set operations



Topic : Derived Relational Algebra operators

- ✓ 1. Intersection ($\cap$)
- * ✓ 2. Join Operations (" $\bowtie$ ")
- * ✓ 3. Division Operation ($\div$)

$$(R \cap S) = R - (R - S)$$





Topic : Projection (π)

- It is used to project the column data from a relation based on the attributes specified with projection operation.

e.g., $\pi_{\langle \text{attribute list} \rangle}(R)$

Attributes/Columns required in the O/p

Name of relation

Projection operator need not obey commutative property
i.e.

$$\pi_{\langle \text{list2} \rangle}(\pi_{\langle \text{list1} \rangle}(R)) \neq \pi_{\langle \text{list1} \rangle}(\pi_{\langle \text{list2} \rangle}(R))$$

(Sid) (Sid, Cid) Sid, Cid Sid

R

Sid	Cid	Branch
S ₁	C ₁	CS
S ₂	C ₂	IT
S ₃	C ₁	CS

①

$$\pi_{\text{Sid, Cid, Branch}}(R) \equiv (R)$$

o/p will be exactly same as relation R.

If projection opⁿ is not present, then all attributes/columns of the relation will be present in o/p

② Retrieve Sids of all the students from relation R.

$$\pi_{\text{Sid}}(R) \Rightarrow \text{o/p} =$$

Sid
S ₁
S ₂
S ₃

③ Retrieve all Cids from the relation R.

$$\pi_{\text{Cid}}(R) \Rightarrow \text{o/p} =$$

Cid
C ₁
C ₂
C ₁

← duplicate tuples are not allowed

R

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS

④

$$\pi_{Sid, Cid}(R) \Rightarrow o/p =$$

Sid	Cid
S_1	C_1
S_2	C_2
S_3	C_1

tuple wise
these are
distinct
tuples.

$$\underbrace{\pi_{Sid}(\pi_{Sid, Cid}(R))}_{\text{it will execute}} \neq \underbrace{\pi_{Sid, Cid}(\pi_{Sid}(R))}_{\text{invalid R.A. query.}}$$



Topic : Selection(σ)

It is used to select the tuples from underlying relation based on the predicate condition specified with selection operation.

$$\sigma_{\langle \text{selection_condition} \rangle}(R)$$

Predicate Condⁿ

Name of relation

NOTE:

$$\sigma_{A \wedge B}(R) = \sigma_{B \wedge A}(R)$$

OR

$$\sigma_B(\sigma_A(R)) = \sigma_A(\sigma_B(R))$$

R

Sid	Cid	Branch
S ₁	C ₁	CS
S ₂	C ₂	IT
S ₃	C ₁	CS

①

$\pi_{\text{Sid, Cid, Branch}}(R)$

We did not specify any selection Condⁿ.
 ∴ All the tuples of the relation
 will be present in o/p

②

Select the records corresponding to Cid = "C₁"

$\sigma_{\text{Cid} = \text{C}_1}(R) =$

Sid	Cid	Branch
S ₁	C ₁	CS
S ₃	C ₁	CS

No projection
 o/p is present.
 ∴ All the attributes will be
 present in o/p

only the tuples in which
 Cid is "C₁" will
 be present in o/p.

R

Sid	Cid	Branch
S ₁	C ₁	CS
S ₂	C ₂	IT
S ₃	C ₁	CS

$\sigma_{Cid=C_1}(R) \Rightarrow \text{o/p} =$

Sid	Cid	Branch
S ₁	C ₁	CS
S ₃	C ₁	CS

③ Select Sids of the students who enrolled for course with Cid = C₁

$\pi_{Sid}(\sigma_{Cid=C_1}(R)) \Rightarrow$

Sid
S ₁
S ₃

R

Sid	Cid	Branch
S ₁	C ₁	CS ×
S ₂	C ₂ ×	IT
S ₃	C ₁	CS ×

④ Retrieve records
in which Cid=C₁
and Branch='IT'

Sid	Cid	Branch

$\sigma_{(Cid=C_1 \wedge Branch='IT')}(R) \Rightarrow$ o/p will be
 empty relation
 Condⁿ $\sigma_{Branch='IT'}(\sigma_{Cid=C_1}(R))$

④ Retrieve records
in which Cid=C₁
or Branch='IT'

$\sigma_{(Cid=C_1 \vee Branch='IT')}(R) \Rightarrow$ o/p =
 Condⁿ

Sid	Cid	Branch
S ₁	C ₁	CS
S ₂	C ₂	IT
S ₃	C ₁	CS



Topic : NOTE



$$\rightarrow \sigma_{A \wedge B}(R) = \sigma_{B \wedge A}(R)$$

OR

$$\sigma_B(\sigma_A(R)) = \sigma_A(\sigma_B(R))$$

OR

$$\sigma_A(R) \cap \sigma_B(R)$$

(OR)

$$\sigma_B(R) \cap \sigma_A(R)$$

$$\sigma_{A \vee B}(R) \equiv \sigma_{B \vee A}(R)$$

(OR)

$$\sigma_A(R) \cup \sigma_B(R)$$

(OR)

$$\sigma_B(R) \cup \sigma_A(R)$$



Topic : Cross Product ($\times$)

/ Cross Join / Cartesian Product .



Cross-product is a binary operation. Let R and S are any two relation, then cross product $R \times S$ will result in all attributes of R followed by all attribute of S with all possible combinations of tuples from R and S.

x attributes

m tuples

R		
Sid	Cid	Branch
S ₁	C ₁	CS
S ₂	C ₂	IT
S ₃	C ₁	CS

y attributes

n tuples

S	
Sid	Sname
S ₁	Ram
S ₂	Mohan
S ₃	Ram

$R \times S =$

$(m \times n)$ tuples

$x+y$ attributes

R.Sid	R.Cid	R.Branch	S.Sid	S.Sname
S ₁	C ₁	CS	S ₁	Ram
S ₁	C ₁	CS	S ₂	Mohan
S ₁	C ₁	CS	S ₃	Ram
S ₂	C ₂	IT	S ₁	Ram
S ₂	C ₂	IT	S ₂	Mohan
S ₂	C ₂	IT	S ₃	Ram
S ₃	C ₁	CS	S ₁	Ram
S ₃	C ₁	CS	S ₂	Mohan
S ₃	C ₁	CS	S ₃	Ram



Topic : Union, Set difference, Intersection

- ❑ Union, Set Difference and Intersection are the Set operations.
- ❑ To use set theory operators on any two relations, those relations must be union compatible.
- ❑ The union compatibility of relations implies that the participating relations must fulfil the following conditions.
 1. Same degree, i.e. The two relations must have the same number of attributes.
 2. Same domain of each corresponding attributes of relations
"data type"

R

Sid	Ad	Branch

No. of attributes
in Relation R $\neq$ No. of attributes
in relation S.

S

Sid	Sname

∴ Given relation R & S are
not Union Compatible.

R

Sid	marks

S

Sid	Sname

① No. of attributes in Relation R = No. of attributes in relation S.

i.e., first condition for union Compatibility is satisfied.

② Domain of first attribute of relation R is same as domain of first attribute of relation S

Because of this R & S are not Union Compatible

{ Both are integer type }
but domain of 2nd attribute of relⁿ R is not same as domain of 2nd attribute of relⁿ S.

R

Sid	Branch

S

Sid	Sname

Given Relation R & S
are Union Compatible.

① No. of attributes in Relation R = No. of attributes in Relation S.

i.e., first Condition for Union Compatibility is satisfied.

② Domain of first attribute of Relation R is same as domain of first attribute of Relation S

{ Both are integer type }

And domain of 2nd attribute of Relⁿ R is also same as domain of 2nd attribute of Relⁿ S.

2nd Condition is also satisfied

Note: ① If relations are Union Compatible, then set operations can be performed on those relation.

and

② After the set operation, the resulting relation will take the names of its attributes from left hand side relation.

ie in " $R \cup S$ ", names of attributes will be same as names of attributes in relation R .

And in " $S \cup R$ "; names of attributes will be same as names of attributes in relation " S ".



Topic : Union, Set difference, Intersection

- ❑ **Union ($A \cup B$)**- It contains unique tuples from both the relations.
- ❑ **Difference ($A - B$)**- It contains all the tuples that are contained in the relation A but are not present in the relation B
- ❑ **Intersection ($A \cap B$)**- It contains all the tuples that are contained in both the relations A as well as in relation B.

Employee (E)

Eid	Ename
1	A
3	B
5	A
8	C

EUS =

Eid	Ename
1	A
3	B
5	A
8	C
2	D
7	C

SNE =

|||
S-(S-E)

Sid	Sname
3	B
5	A

Student (S)

Sid	Sname
2	D
3	B
5	A
7	C

S-E =

Sid	Sname
2	D
7	C

E-S =

Eid	Ename
1	A
8	C



Topic : Rename (ρ)

Student(Student-id, Student-name)



Rename operation can be used to rename attribute of the relation, name of the relation or both.

* Renaming a relation:

$\rho_{\text{FinalYrStudents}}(\text{Students})$

Rename op

Current name of relation

Names of attributes will remain same

New name assigned to relation

* Renaming attributes:

$\rho_{(\text{SID}, \text{Sname})}(\text{Students})$

Current name of relation

Name of relation will remain same

New name for first attribute

New name for second attribute

* Renaming both:

$\rho_{\text{FinalYrStudents}}(\text{SID}, \text{Sname})(\text{Students})$

New Name for relation

Current name of relation



Topic : Derived Relational Algebra operators

1. Intersection ($\cap$) $\rightarrow$ It is a set opⁿ. $\therefore$ Intersection of two relations can be performed only if relations are Union Compatible
- ✓ 2. Join Operations (" $\bowtie$ ")
 - $\rightarrow$ It is a derived relational algebra opⁿ, which is derived using "Set difference" opⁿ
ie, $R \cap S = R - (R - S)$
3. Division Operation ($\div$)



Topic : Join Operations

- * Join operations are used to join relational tables based on some condition.
- * If we do not specify any Condⁿ, then it becomes Cross Join (i.e., Cross Product)
- * In cross join, all the tuples of 1st relation are "joined" with all the tuples of 2nd relation {i.e., $R \times S$ }



Topic : Types Join Operations

- ✓ 1) Inner join
 - ✓ a. Theta join
 - ✓ b. Equi join
 - ✓ c. Natural join

{ In inner join, we select only those tuples from both the relation that satisfy join Condⁿ }

- ✓ 2) Outer join
 - ✓ a. Left outer join
 - ✓ b. Right outer join
 - ✓ c. Full outer join

{ In outer join we may also select the tuples that did not satisfy join Condⁿ }



Topic : Inner join



- ▶ Inner join, includes only those tuples that satisfy the matching criteria.



Topic : Theta join

- The general case of JOIN operation is called a Theta join. It is denoted by symbol θ *i.e., $\bowtie_{\theta}$ it may represent any condition*

Theta join can use any conditions in the selection criteria.

$<, \leq, >, \geq, <>, =$

$R_1 \bowtie_{\theta} R_2$
Condⁿ

Student (S)

Sid	Sname
2	A
3	B
4	A

Employee (E)

Eid	Ename
2	B
4	A
5	B

Student ⋈ Employee
 (Student.Sid < Employee.Eid)

It is the join condn 'θ'
 it can be anything
 %p =

$$\equiv \left(\sigma_{S.Sid < E.Eid} (S \times E) \right)$$

S.Sid	S.Sname	E.Eid	E.Ename
2	A	5	B
3	B	4	A
4	A	2	B



Topic : Equijoin



{ It is also a type of θ join, }
{ but in Equijoin Condⁿ is Equality Condⁿ }

An equijoin is a theta join using the equality operator.
In equijoin the condition is always equality condition.

It is a type of theta join.

Need not be on
Common attributes of
relation.

{ If at all it is on
Common attributes, then
also need not be on
all Common attributes }

Student (S)

Sid	Sname
2	A
3	B
4	A

Employee (E)

Eid	Ename
2	B
4	A
5	B

Student ⋈

(Student.Sid = Employee.Eid)

Attribute need not be
Common attribute

Employee

Condⁿ for join
is Equality Condⁿ.
∴ Equi-join

≡

$\left(\sigma_{S.Sid = E.Eid} (S \times E) \right)$

∴ p =

S.Sid	S.Sname	E.Eid	E.Ename
2	A	2	B
4	A	4	A

Student (S)

Sid	Sname
2	A
3	B
4	A

Enroll (E)

Sid	Cid
2	C ₁
4	C ₂
5	C ₃

Student ⋈

(Student.Sid = Enroll.Sid)

Attribut may Common attribute as well

Enroll

Condⁿ for join is Equality Condⁿ,
i.e. Equi-join

≡

$\sigma_{S.Sid = E.Sid} (S \times E)$

O/p :

S.Sid	S.Sname	E.Sid	E.Cid
2	A	2	C ₁
4	A	4	C ₂

When we use Equijoin, Common attributes for both the relations will be present in o/p

Note:-

If we use theta join or Equijoin then we need to specify the join Condition Explicitly, which is not the case with Natural join

{ In Natural join, join Condⁿ is by default }
Equality Condⁿ On all common attributes }



Topic : Natural join ($\bowtie$)

Natural join is performed based on the equality condition on all common attributes (column) between the relations. The name and type of common attributes must be same.

Additionally, a natural join removes the duplicate columns involved in the equality comparison so only one of each compared column remains

Consider relⁿ, Student (Sid, Sname) & Enroll (Stud-id, Course-id)

There is no common attribute b/w Student & Enroll
o Sid & Can not be compared using Natural join
But we can still use Equijoin.

Natural Join ($\bowtie$)

Natural join is a derived relational algebra opⁿ.
which can be derived using three basic relational algebra opⁿ.
i.e; cross product, Selection and projection.

Derivation of Natural join

$R \bowtie S =$

Step-1 : obtain $R \times S$

Step-2 : Select the tuples from $R \times S$ based on Equality Condⁿ on all common attributes of R and S .

Step-3 : From the output of Step-2 project distinct attributes

Student (S)

Sid	Sname
2	A
3	B
4	A

Enroll (E)

Sid	Cid
2	C ₁
4	C ₂
5	C ₃

$$\text{Student} \bowtie \text{Enroll} = \pi_{\text{Sid, Sname, Cid}} \left(\sigma_{\text{Student.Sid} = \text{Enroll.Sid}} (\text{Student} \times \text{Enroll}) \right)$$

No condn is specified,
so it is Natural join

O/p =

"Sid" will be present only once in the o/p

Sid	Sname	Cid
2	A	C ₁
4	A	C ₂

eg: $R(A\ B\ C)$ & $S(B\ C\ D)$

$$R \bowtie S = \pi_{A,B,C,D} \left(\left\{ \begin{array}{l} R.B = S.B \\ \wedge \\ R.C = S.C \end{array} \right\} (R \times S) \right)$$

Equality Condⁿ
on all common
attributes

eg. $R(A, B, C)$ & $S(D, E)$

$$R \bowtie S = \pi_{A, B, C, D, E}(\underbrace{(R \times S)}_{\text{No common attributes b/w R \& S}}) \equiv R \times S$$

No common attributes b/w R & S
∴ No selection Condⁿ.

And hence all the tuples of $R \times S$
will get selected

And finally all attributes
of $R \times S$ will be projected.

∴ O/p will be exactly same as $R \times S$

If there is no common attribute between R and S, then Natural join of R and S will be equal to " $R \times S$ ".

Student (S)

Sid	Sname
2	A
3	B
4	A

Employee (E)

Eid	Ename
2	B
4	A
5	B

$$\text{Student} \bowtie \text{Enroll} \equiv R \times S$$

↑
No Common
attributes b/w
Student & Enroll



Topic : Outer join

In an outer join, along with tuples that satisfy the matching criteria, we also include tuples that do not match the criteria either from left hand side relation, or from right hand side relation or both.



Topic : Left Outer Join ($\bowtie$)



$R \bowtie S =$ All tuples of $R \bowtie S$ along with the tuples from left hand side relation { i.e, $R \in R$ } that failed the join Condⁿ are present.



Topic : Right Outer Join ($\bowtie$)

$R \bowtie S =$ All tuples of $R \bowtie S$ along with the tuples of right hand side relation {i.e., $R \bowtie S$ } that failed the join condition are present



Topic : Full Outer Join ($\bowtie$)

$R \bowtie S$: All tuples of $R \bowtie S$ along with the tuples from both side relation that failed the join condition are present.

R:-

A	B	C
1	2	3
1	4	5

RMS=

A	B	C	D
1	2	3	4

S:-

B	C	D
2	3	5
5	6	7
7	7	9

RMS=

A	B	C	D
1	2	3	4
1	4	5	NULL

RMS=

A	B	C	D
1	2	3	4
NULL	5	6	7
NULL	4	7	9

RMS=

A	B	C	D
1	2	3	4
1	5	6	NULL
NULL	4	7	7
NULL	4	7	9

118: Result of which among the following is a super set of other three

(a) Natural join

(b) Left outer join

(c) Right outer join

☒ (d) Full outer join



Topic : Division ($\div$)



Division operation is used whenever the query is with respect to every or all.



2 mins Summary



Topic

Derived relational algebra operations

Topic

Different types of join operations



Topic

Division operation

THANK - YOU